

Suprathika
Email: suprathika.diyala9@gmail.com
Phone: +1 (551) 364-2251

PROFESSIONAL SUMMARY:

- 5+ years of industry experience with solid understanding of Data Modeling, Evaluating Data Sources, and strong
- understanding of Data Warehouse/Data Mart Design, Data Mining ETL, BI, Data visualization, Data Validation
- OLAP, Client/Server applications.
- Optimized predictive models for Telefonica and Bouygues Telecom, boosting customer retention by \$5M/month using advanced machine learning techniques.
- Automated feature engineering for 400,000+ customer datasets, exceeding the monthly revenue target by 133%,
- utilizing Python and SQL for efficient data processing.
- Developed real-time model monitoring using PostgreSQL, Python, and R, enhancing data-driven
- decision-making with a focus on data visualization and model performance tracking.
- Enhanced data preprocessing techniques, reducing noise and improving predictive accuracy by 6% through
- advanced data wrangling and ETL practices.
- Developed a single attribution-based churn metric, sustaining a 2% incremental gain for five months, leveraging
- statistical analysis and churn prediction models.
- Performed data wrangling on a stream averaging 30,000 calls daily to identify patterns in changing customer
- behavior, applying clustering and time series analysis to uncover insights.
- Expertise in utilizing **Amazon EC2** to compute query processing and storage in diverse applications.
- Expertise in using **AWS S3** for data staging, transfer, and archiving.
- Experience with **AWS Redshift** for large-scale data migrations and **CDC** implementation.
- Extensive experience with NoSQL databases and Hadoop cluster integration, including **HBase, Cassandra, MongoDB, DynamoDB, and CosmosDB**.
- Experience designing scalable and resilient **Restful APIs, ETL** solutions, and third-party platform integrations for the Enterprise Site platform.
- Experience in implementing pipelines using **ELK(Elasticsearch, logstash, kibana)** and developing stream processes using **Apache Kafka**.
- Experience in working with various streaming ingest services with Batch and Real-time processing using **Spark Streaming, Kafka, Confluent, Storm, Flume, and Sqoop**.
- Good experience in the development of **Bash scripting, T-SQL, and PL/SQL Scripts**.
- Experience with build automation and continuous integration tools such as **Apache ANT, Maven, and Jenkins**.
- Experience in implementation of web analytics tools such as **Heap, Adobe Analytics** and marketing tools **MailChimp**.
- **Designed, trained, and optimized Generative AI models** (including GPT, DALL-E) to automate content creation, significantly improving operational efficiency and reducing manual effort.
- Delivered enterprise level projects under budget and on time to Fortune 500 clients with global cross functional teams.

TOOLS AND TECHNOLOGIES:

Databases/ETL/Query	Teradata, SQL Server, Postgres and Hadoop (Map Reduce); SQL, AWS S3, EMR, EC2 Hive, Pig and Alteryx.
Visualization	Tableau, ggplot2 and R-Shiny
Statistics	Exploratory Data Analysis: Univariate/Multivariate Outlier detection, Missing value imputation, Histograms/Density estimation, EDA in Tableau. Supervised Learning: Linear/Logistic Regression, Lasso, Ridge, Elastic Nets, Decision Trees, Ensemble Methods, Random Forests, Support Vector Machines, Gradient Boosting, Neural Networks. Unsupervised Learning: Principal Component Analysis, Association Rules, Factor Analysis, K-Means, Hierarchical Clustering, Market Basket Analysis.
Machine Learning	R: caret, glmnet, forecast, xgboost, rpart, survival, arules, sqldf, dplyr, nloptr, lpSolve, ggplot. Python: pandas, numpy, scikit-learn, scipy, stats models, matplotlib, tensorflow. SAS: Forecast server, SAS Procedures and Data Steps. Spark: MLlib, GraphX. SQL: Subqueries, joins, DDL/DML statements.

PROFESSIONAL EXPERIENCE:

Client: Scripps, San Diego, CA

July 2024 – Present

Role: Data Scientist

Responsibilities:

- Spearheaded the development of a probabilistic classification algorithm, leveraging Bayesian statistics to compute conditional probabilities, boosting auction success rate by 20% and predicting a 3% revenue increase.
- Conducted advanced statistical analysis and predictive modeling to provide high-level insights, preventing a PKR 40 million loss for senior management.
- Applied p-values and correlation coefficients for feature selection, identifying key auction predictors.
- Leveraged Power BI for data visualization and storytelling, driving strategic decisions.
- Worked closely with the data engineering team to ensure the seamless integration of NPS and guest metrics data into a central data warehouse for streamlined reporting.
- Presented data-driven insights to stakeholders and senior leadership, influencing strategic decisions to improve customer satisfaction and enhance operational efficiencies.

- Configured Adobe Analytics for advanced event tracking, collaborating with development teams to implement Adobe Launch for tag management across digital platforms.
- Partnered with marketing teams to analyze A/B testing results using Adobe Analytics, leading to data-driven decisions on content and feature optimizations.
- Conducted hands-on validations of financial risk models, ensuring compliance with internal audit requirements and regulatory deliverables.
- Validated and improved machine learning models for healthcare datasets, focusing on accuracy, bias identification, and data consistency.
- Effectively communicated data analysis findings to both technical and non-technical stakeholders, ensuring alignment on strategic healthcare initiatives.
- Deep understanding of healthcare coding systems and medical terminology used for data validation and compliance.
- Executed data strategy initiatives, focusing on governance and quality to empower the FRM area.
- Conducted cross-channel analytics, integrating Adobe Analytics data with other digital platforms like Google Analytics to provide a holistic view of marketing efforts and ROI.
- Updated Python scripts to match training data with our database stored in AWS Cloud Search, so that we would be able to assign each document a response label for further classification.
- Developed and maintained data processing applications using Spark and Scala, optimizing data workflows and ensuring high-performance execution.
- Utilized test-driven development (TDD) practices to ensure the reliability and stability of Scala-based Spark applications in production environments.
- Expertise in quantitative analysis, data mining techniques, and the presentation of data to see beyond the numbers and understand trends and insights.
- Creating complex SQL queries and scripts to extract and aggregate data to validate the accuracy of the data and Business requirement gathering and translating them into clear and concise specifications and queries.
- Extracted data using SQL from data sources and performed Exploratory Data Analysis (EDA) and Data Visualizations using R, and Tableau.
- Implemented Uni-variate and Bi-variate analysis to understand the intrinsic/combined effects.
- Applied concepts of probability, distribution, and statistical inference on the given dataset to unearth interesting findings using comparison, T-test, F-test, R-squared, P-value etc.
- Developed and applied predictive models using multiple statistical and machine learning techniques. Solid understanding and working knowledge of statistical modeling techniques such as linear regression, logistic regression, generalized linear models, data visualization, Random Forest, text mining, tree models, cluster analysis, principal components and feature creation.

Environment: R - (R studio), Python - (Anaconda), SQL Server Management System, Tableau, Excel.

Client: GDS Telecom, Atlanta

October 2023 –June 2024

Role: Data Scientist

Responsibilities:

- Analyzed product plans in competitive markets using A/B testing and recommended metrics to increase operational efficiency with model development using DataRobot.
- Implemented Data pipelines for big data processing using Spark Clusters in Amazon EMR, Python and performed customer segmentation.
- Implemented business processing models using predictive & prescriptive analytics on transactional data with regression, classification.

- Designed workflows using SQL queries for data preprocessing such as cleansing and transformation of data to implement & automate KPI dashboards.
- Implemented Logistic, Random forests with Python packages to decide insurance purchase by a USAA member.
- Developed and managed end-to-end AI pipelines for Generative models, ensuring data quality, model performance, and scalability for large datasets.
- Streamlined data collection and pre-processing pipelines for training Generative AI models, improving data efficiency and model training time.
- Utilized Snowflake for building data warehousing solutions, including data migration, data modeling, and real-time analytics for large datasets, ensuring performance and scalability.
- Developed and deployed machine learning models including XGBoost, DB Scan, Gaussian Mixture Models for classification, clustering, and regression tasks.
- Developed and maintained robust API connections for data ingestion and transformation involving EDI 837 and 835 standards, ensuring compliance with healthcare data regulations.
- Implemented OpenCV techniques to extract text and relevant data points from scanned medical forms and documents to automate data entry.
- Developed scripts using PyAutoGUI to interact with desktop applications and automate the processing of healthcare data files.
- Automated web interactions for data extraction from insurance portals and verification sites using Selenium, enhancing the speed and reliability of data gathering.
- Built and maintained scalable data pipelines that processed large volumes of healthcare data, leveraging Python and Spark to support real-time claims processing and analytics.
- Enhanced data pipelines to include automated error handling and logging, ensuring uninterrupted data flow and compliance with healthcare regulations.
- Applied machine learning algorithms to develop predictive models that forecast claim approval probabilities, helping reduce claim denial rates by 15%.
- Designed supervised learning models to analyze patterns in healthcare payment data, aiding in proactive denials management.
- Deployed automation solutions using Azure cloud platforms, ensuring scalability and reliability through containerized applications using Docker and Kubernetes.
- Integrated cloud-native services for optimized storage and processing of healthcare data, facilitating efficient data management and retrieval.
- Improved model accuracy through hyperparameter tuning and model optimization.
- Ensured data quality and regulatory compliance in healthcare projects by developing thorough validation procedures and performing data checks.
- Reviewed and flagged inaccuracies in datasets, ensuring consistency and factual correctness in healthcare-related data.
- Led the development of scalable analytical solutions to track NPS trends and customer behavior over time, enabling predictive insights.
- Collaborated with model owners and developers to identify model limitations, performing ad hoc analyses and recommending effective remediations.
- Developed and maintained comprehensive data quality performance metrics, enhancing reporting for senior management.
- Applied advanced data visualization techniques in Power BI/Tableau to create user-friendly NPS dashboards, providing clear insights into customer feedback and operational performance.
- Implemented HDFS clusters from unstructured marketing data platforms with HiveQL, Spark, Hadoop technologies.

- Collaborated with business owners of products for understanding business needs and automated business processes and data storytelling in Tableau.
- Used pandas, numpy, Seaborn, scipy, matplotlib, sci-kit-learn in Python for developing various machine learning algorithms.
- Established monitoring and alerting systems using Stackdriver to proactively manage the health and performance of GCP services and data pipelines.
- Collaborated with cross-functional teams to gather data requirements and design scalable solutions, improving business processes and decision-making.
- Deployed artifacts from lower to higher environments, ensuring smooth transitions and continuity of data processing operations.
- Leveraged Adobe Analytics to assess and track the performance of digital marketing campaigns, optimizing strategies based on real-time data and performance metrics.
- Integrated Adobe Analytics with data visualization tools like Tableau and Power BI to automate performance reporting, delivering timely insights to executives and marketing teams.
- Executed process improvements in data workflows using Alteryx processing engine.
- Worked as Data Architects and IT Architects to understand the movement of data and its storage.
- Data Manipulation and Aggregation from different source using Nexus, Toad, Business Objects and SmartView.
- Implemented Agile Methodology for building the applications and data framework development.
- Focus on integration overlap and Informatica newer commitment to MDM with the acquisition of Identity Systems.
- Delivered various complex OLAP databases/cubes, scorecards, dashboards and reports.
- Handled importing data from various data sources, performed transformations using Hive, MapReduce, and loaded data into HDFS.
- Researched, evaluated, architected, and deployed new tools, frameworks, and patterns to build sustainable Big Data platforms for the clients.

Environment: R, Python, TensorFlow, Machine Learning, Tableau, Bigdata, Alteryx, Hive, OLAP, DB2, Metadata, MS Excel, DataRobot.

Client: HDFC, India.

April 2020 –Sept 2023

Role: Data Scientist

Responsibilities:

- Developed real-time model monitoring using PostgreSQL, Python, and R, enhancing data-driven decision-making with a focus on data visualization and model performance tracking.
- Enhanced data preprocessing techniques, reducing noise and improving predictive accuracy by 6% through advanced data wrangling and ETL practices.
- Developed a single attribution-based churn metric, sustaining a 2% incremental gain for five months, leveraging statistical analysis and churn prediction models.
- Performed data wrangling on a stream averaging 30,000 calls daily to identify patterns in changing customer behavior, applying clustering and time series analysis to uncover insights. Implemented Generative AI-powered chatbots and virtual assistants, improving customer support efficiency and reducing response time.
- Proficient in writing advanced SQL queries involving Window Functions, Group By, Joins, Partitions, and Indexing to optimize data retrieval and processing for large-scale datasets.
- Working with Parquet files and Impala using PySpark, and Spark Streaming with RDDs and Data Frames.

- Aggregating logs data from various servers and made them available in downstream systems for analytics by using Apache Kafka. Improved Kafka performance and implemented security.
- Developing batch and streaming processing apps using Spark APIs for functional pipeline requirements.
- Automating data storage from streaming sources to AWS data lakes like S3, Redshift and RDS by configuring AWS Kinesis (Data Firehose).
- Performing analytics using real time integration capabilities of AWS Kinesis (Data Streams) on streamed data.
- Cleaning and handling missing values in data using Python by backward-forward filling methods and applied Feature engineering, normalizing and label encoding techniques using Python Scikit-learn preprocessing.
- Ensuring database performance in production by stress testing AWS EC2 of DynamoDB environments.
- Automating deployments and routine tasks using UNIX Shell Scripting.
- Collaborating with the Data Science team building machine learning models on Spark EMR cluster to deliver the data needs under business requirements.
- Working in an agile environment to implement projects and enhancements with weekly SCRUMs.

Environment: Hadoop 2.x, Spark v2.0.2, Hive, Sqoop, Kafka, Spark streaming, ETL, tdd, Python (Pandas, Numpy), PySpark, GIT (version control), MySQL, MS SQL, MongoDB, AWS (EC2, S3, EMR, RDS, Lambda, Kinesis, Redshift, Cloud Formation).